Trump Tweets Retrieval System (RAG)

Hybrid RAG for Political Text Analysis

01

AI - Research seminar

Project Motivation

Modern information retrieval systems must understand the meaning of user queries, not only keywords.
Political communication on social media generates large volumes of short text messages. Searching through such data using traditional keyword search is often inefficient.

Our idea was to build an AI-powered retrieval system that allows users to ask natural language questions and retrieve relevant tweets from Donald Trump's Twitter archive.

The project demonstrates how Hybrid RAG architectures improve the quality of search results.

Project vision and mission

The goal of the project is to develop an AI-based retrieval system for analyzing political tweets using modern natural language processing techniques.

01.

Project goal

Build a system that retrieves relevant tweets using semantic search and hybrid retrieval techniques.

02.

Data set

Tweets from the dataset “Trump Tweets (2009–2025)” collected from Kaggle.

03.

Expected results

An interactive system capable of answering user questions and retrieving relevant tweets.

Dataset and Data Processing

The dataset used in this project is “Trump Tweets (2009–2025)” from Kaggle.

It contains social media posts written by Donald J. Trump on Twitter (X) and Truth Social.

Dataset characteristics:
- Time period: May 2009 – January 2026
- Thousands of tweets and posts
- Political communication and public statements

Source: Kaggle – DataDrivenDecisionLab: Trump Tweets (2009-2025)

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AI Models and Tools

The system integrates several modern AI models and tools.

- Embedding model
- BAAI bge-m3

- Reranking model
- cross-encoder/ms-marco-MiniLM-L-6-v2

- Generation model
- arcee-ai/trinity-large

The system was implemented in Python and deployed using Streamlit and HuggingFace Spaces.

13536	2022-04-29	Truth Social	Thank you to all …	https://truthsoci…
13537	2022-04-28	Truth Social	I'M BACK! #COVFEFE	https://truthsoci…
13538	2022-02-14	Truth Social	Get Ready! Your f…	https://truthsoci…
13539	2021-01-08	Twitter	To all of those w…	https://x.com/rea…
13540	2021-01-08	Twitter	The 75,000,000 gr…	https://x.com/rea…
13541	2021-01-06	Twitter	These are the thi…	https://x.com/rea…
13542	2021-01-06	Twitter	I am asking for e…	https://x.com/rea…
13543	2021-01-06	Twitter	Please support ou…	https://x.com/rea…
13544	2021-01-06	Twitter	Mike Pence didn't…	https://x.com/rea…
13545	2021-01-06	Twitter	These scoundrels …	https://x.com/rea…
13546	2021-01-06	Twitter	Even Mexico uses …	https://x.com/rea…
13547	2021-01-06			

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Hybrid Retrieval Pipeline

01 Query Input
User enters a natural language question.

02 BGE-M3 Dense
Semantic similarity search retrieves candidate tweets.

03 BM25 Search
Keyword-based retrieval improves coverage.

04 Fusion + Dedupe
Results from multiple retrieval methods are merged.

05 Reranking
Cross-encoder reranks tweets based on relevance.

06 Top-K Results
Most relevant tweets are returned to the user.

System Results

The hybrid retrieval approach improves the relevance of search results.
By combining semantic embeddings and keyword search, the system retrieves tweets related to complex political questions.
The reranking model further improves the quality of final results by prioritizing the most relevant tweets.
This approach demonstrates the effectiveness of hybrid retrieval systems for analyzing political text datasets.

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07

SWOT

Weaknesses

- LLM inference requires significant computational resources
- Retrieval quality depends on dataset coverage and preprocessing quality

Threats

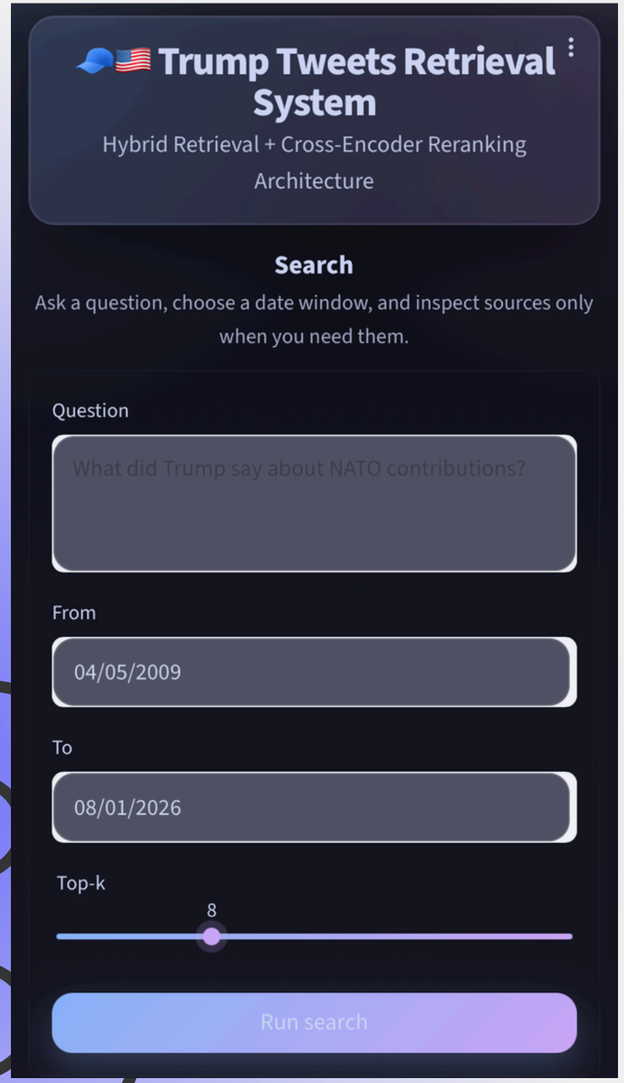
- Political datasets may contain biased or controversial content.
- Misinformation in source data can influence generated answers.

Strengths

- Hybrid retrieval improves relevance by combining semantic and keyword search.
- RAG reduces hallucinations by grounding answers in real tweets.

Opportunities

- System can be extended with sentiment analysis or topic modeling.
- Larger political datasets could improve retrieval coverage.


🧢🇺🇸 Trump Tweets Retrieval System
Hybrid Retrieval + Cross-Encoder Reranking Architecture
Search
Ask a question, choose a date window, and inspect sources only when you need them.
Question
What did Trump say about NATO contributions?
From
04/05/2009
To
08/01/2026
Top-k
8
Run search

SCAN